Fiscal Year:	FY 2008	**Task Last Updated:**	FY 02/01/2008
PI Name:	Bloomberg, Jacob J. Ph.D.		
Project Title:	Development of a Gait Adaptability Training Program as a Countermeasure for Postflight Locomotor Dysfunction		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Sensorimotor Adaptation Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jacob.j.bloomberg@nasa.gov	**Fax:**	FY 281-244-5734
PI Organization Type:	NASA CENTER	**Phone:**	281-483-0436
Organization Name:	NASA Johnson Space Center		
PI Address 1:	NASA Emeritus Scientist, Biomedical Research and Environmental Sciences Div		
PI Address 2:	2101 NASA Parkway, SK272		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058-3607	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	10/01/2004	**End Date:**	10/01/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 10/1/2008 because of reporting requirements (3/09 jvp)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Cohen, Helen (Baylor College of Medicine) Mulavara, Ajitkumar (USRA)		
Grant/Contract No.:	NCC 9-58-NA00403		
Performance Goal No.:			
Performance Goal Text:			

Astronauts experience disturbances in sensorimotor function following their return to Earth due to adaptive responses that occur during exposure to the microgravity conditions of space flight. During exploration class missions these sensorimotor disturbances may lead to disruption in the ability to ambulate and perform functional tasks during the initial introduction to a novel gravitational environment. At present no operational countermeasure has been implemented to mitigate the risks associated with the sensorimotor disturbances that occur during and following space flight. Therefore, the goal of this project is to develop both preflight and in-flight sensorimotor adaptability training programs that will facilitate recovery of locomotor and functional capabilities when astronauts transition to different gravitational environments. Information obtained from this study will enable us to optimize a sensorimotor adaptability training program. The results from this study will also be used to specify the requirements for future in-flight treadmills and other exercise devices used during space flight. Data obtained from this study will also help in the development of preflight sensorimotor adaptability training programs that can be used to support future space flight missions.
Strong evidence in both the motor learning and clinical rehabilitation literature shows that training with task variability along with exposure to repeated sensorimotor adaptive challenges leads to faster adaptation to new environments and readaptation to the normal environment. By applying these concepts for training astronauts we can enhance their ability to “learn how to learn” to adapt to new gravitational environments. Sensorimotor adaptability training can be integrated with existing treadmill exercise programs. The training program can be performed simultaneously with nominal inflight treadmill exercise for long-duration flights on the ISS that simulate Mars transit times but can be implemented in a preflight training mode to support lunar missions. By manipulating the sensory conditions of exercise by varying visual flow patterns, body load, and the stability of the walking surface the training systematically and repeatedly challenges the balance control system during treadmill walking. This training regimen enhances sensorimotor adaptability serving to facilitate adaptation to gravitational environments.

To develop the training program we are conducting a series of ground-based studies that systematically investigate the training efficacy associated with exposing subjects to different combinations of sensory input during treadmill locomotion including alterations in visual flow, body loading, and support surface stability. We are also determining the optimal training mode (strategic vs. adaptive) and retention characteristics to maximize the efficacy, efficiency of the training procedure.

Task Description:

In the third year we have had three major accomplishments:

1) We demonstrated that sensorimotor adaptability training on a treadmill, using alterations in visual input as the training modality, confers increased adaptability in overground ambulation on an obstacle course while experiencing a novel visual distortion. Therefore, adaptability training performed on a treadmill does not have limited applicability but transfers to more complex and functional behaviors.

2) We have completed a study examining the use of body unloading as a gait adaptability training modality. The objective of this study was to investigate if limited exposure to unloaded treadmill walking leads to adaptive alteration in locomotor function. Subject’s locomotor function was tested before and after 30 minutes of treadmill walking at 40% body weight support produced by a pneumatically controlled harness device integrated into the treadmill system. Data indicate that following unloaded locomotion there were adaptive changes in vestibularly mediated head movement control systems, lower limb kinematics and gait cycle timing. This study is the first to demonstrate that the body load sensing system can be adaptively modified following brief exposures to unloaded locomotion. From a mechanistic standpoint we have shown that afferent sensory information regarding body load can adaptively modulate vestibular activity. This has implications for our understanding of the sensory integration processes that contribute to the control of motor output.

3) We completed hardware integration and NASA safety reviews for our treadmill-motion based device. We now have the capability of producing support surface variation during gait training by using a treadmill mounted to a six-degree-of-freedom motion device. We have completed a study examining how different modes of support surface movement during locomotion can serve as a training modality. The objective of this study was to investigate if limited exposure to various types of unstable walking surface configurations induces adaptive alterations in locomotor function. Currently, analysis of data from this study is underway. Preliminary results indicate that subjects can be trained and can adapt to a variety of support surface variations.

Plans for next year include completion of a comprehensive training study designed to investigate the efficacy and retention characteristics of visual flow, unstable surface and combined visual flow/unstable support surface training.

Rationale for HRP Directed Research:

From a clinical perspective exposure to varied sensorimotor stimuli during treadmill walking provides environmental enrichment to enhance gait and postural adaptability. A vast literature demonstrates that the central nervous system requires a rich and varied sensorimotor environment to maintain normal structure and function. Also, a converging body of evidence supports the use of environmental enrichment both before and after brain trauma, as an intervention to help restore behavioral function These studies indicate that exposure to environmental enrichment can play a central role in the restoration of functional mobility. Importantly, studies have shown that training comprised of challenging balance tasks compared to simple repetitive or treadmill exercises alone increase structural plasticity in the motor cortex and thalamus.
Within the context of locomotor functional training and rehabilitation, it is clear that an enriched and challenging environment provided by virtual reality devices provides the optimal conditions to explore variability and in that exploration increase adaptability of motor function. Indeed, the concept of employing virtual reality technology as a tool for rehabilitation has become a topic of great interest among a number of researchers. A collateral benefit of the application virtual reality technology, in this context, will be to make gait training programs for both space flight and rehabilitation more interesting, ultimately leading to increased adherence to prescribed training regimens.

Research Impact/Earth Benefits:

In addition, the gait adaptability training program being developed by this project can be used to reduce the propensity of falling in the elderly. We have pursued this line of research in collaboration with Drs. Ronita Cromwell and Regina Buccello-Stout at the University of Texas Medical Branch in Galveston. This associated study compared a group of elders receiving the specialized treadmill adaptability training developed under NSBRI funding with a group that received the treadmill training only. Results from this study showed significantly improved gait function in elders that received the sensorimotor adaptability training. This study clearly demonstrated the general utility of the training method beyond the NASA applications.

Task Progress:	In the third year we have had three major accomplishments: 1) We completed a study showing that sensorimotor adaptability training on a treadmill, using alterations in visual input as the training modality, confers increased adaptability in overground ambulation on an obstacle course while experiencing a novel visual distortion. Therefore, adaptability training performed on a treadmill does not have limited applicability but transfers to more complex and functional behaviors. Two manuscripts have been submitted describing the results of this study. 2) We completed a study examining the use of body unloading as a gait adaptability training modality. The objective of this study was to investigate if limited exposure to unloaded treadmill walking leads to adaptive alteration in locomotor function. Subject's locomotor function was tested before and after 30 minutes of treadmill walking at 40% body weight support produced by a pneumatically controlled harness device integrated into the treadmill system. Data indicate that following unloaded locomotion there were adaptive changes in vestibularly mediated head movement control systems, lower limb kinematics and gait cycle timing. This study is the first to demonstrate that the body load sensing system can be adaptively modified following brief exposures to unloaded locomotion. From a mechanistic standpoint we have shown that afferent sensory information regarding body load can adaptively modulate vestibular activity. This has implications for our understanding of the sensory integration processes that contribute to the control of motor output. 3) We completed hardware integration and NASA safety reviews for our treadmill-motion based device. We now have the capability of producing support surface variation during gait training by using a treadmill mounted to a six-degree-of-freedom motion device. We have completed a study examining how different modes of support surface movement during locomotion can serve as a training modality. The objective of this study was to investigate if limited exposure to various types of unstable walking surface configurations induces adaptive alterations in locomotor function. Currently, analysis of data from this study is underway. Preliminary results indicate that subjects can be trained and can adapt to a variety of support surface variations.
Bibliography Type:	Description: (Last Updated: 11/14/2025)
Articles in Peer-reviewed Journals	Cohen HS, Bloomberg JJ. "Modified dynamic visual acuity tests after acoustic neuroma resection." Acta Otolaryngol. 2007 Aug;127(8):825-8. PMID: 17762992 , Aug-2007
Articles in Peer-reviewed Journals	Cohen HS, Mulavara AP, Bloomberg JJ. "Importance of critical features of training in promoting adaptive generalization of locomotor function." Gait and Posture. Submitted, 2007. , Oct-2007
Articles in Peer-reviewed Journals	Mulavara AP, Houser J, Miller C, Bloomberg JJ. "Full-body gaze control mechanisms elicited during locomotion: effects of VOR adaptation." J Vestib Res. 2005;15(5-6):279-89. PMID: 16614474 , Oct-2005
Articles in Peer-reviewed Journals	Peters BT, Bloomberg JJ. "Dynamic visual acuity using "far" and "near" targets." Acta Otolaryngol. 2005 Apr;125(4):353-7. PMID: 15823804 , Apr-2005
Articles in Peer-reviewed Journals	Roller CA, Cohen HS, Bloomberg JJ, Mulavara AP. "Variable practice improves obstacle avoidance on a compliant surface." Perceptual Motor Skills. Submitted, 2007. , Oct-2007
Articles in Peer-reviewed Journals	Moore ST, MacDougall HG, Peters BT, Bloomberg JJ, Curthoys IS, Cohen HS. "Modeling locomotor dysfunction following spaceflight with Galvanic vestibular stimulation." Exp Brain Res. 2006 Oct;174(4):647-59. PMID: 16763834 , Oct-2006
Articles in Peer-reviewed Journals	Mulavara AP, Richards JT, Ruttley T, Marshburn A, Nomura Y, Bloomberg JJ. "Exposure to a rotating virtual environment during treadmill locomotion causes adaptation in heading direction." Exp Brain Res. 2005 Oct;166(2):210-9. PMID: 16034569 , Oct-2005
Articles in Peer-reviewed Journals	Nomura Y, Mulavara AP, Richards JT, Brady R, Bloomberg JJ. "Optic flow dominates visual scene polarity in causing adaptive modification of locomotor trajectory." Brain Res Cogn Brain Res. 2005 Dec;25(3):624-31. PMID: 16216478 , Dec-2005
Articles in Peer-reviewed Journals	Richards JT, Mulavara AP, Bloomberg JJ. "The interplay between strategic and adaptive control mechanisms in plastic recalibration of locomotor function." Exp Brain Res. 2007 Apr;178(3):326-38. PMID: 17061092 , Apr-2007